# Capstone Project Weekly Report

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| **Reporting Week:** | Week ending October 11, 2026 | **Sponsor:** | Deloitte |
| **Project Title:** | Air Conditioning Demand Forecasting & Supply Chain Optimization | **Report Owner:** | Sai Prashanth Sarathy / R Arvindshankar (PGID: 12510081 / 12510053) |
| **Team Members:** | Arvind, Hariharan, Vani, Vedika, Sai Prashanth | **Program:** | ISB AMPBA (Summer 2026) |

## Project Milestones

The team achieved complete technical and operational completion of the enterprise Demand Intelligence & Supply Chain Optimization platform. We finalized Hierarchical Multi-Echelon aggregation and Bottom-Up / Top-Down reconciliation (src/hierarchy/), implemented an advanced Explainability & Interpretability Suite (src/explainability/shap_analysis.py), deployed the production-grade application to Vercel with 100% localhost parity and APICS replenishment smoothing telemetry, completed an internal code and methodology review (review_result.md) backed by the automated test suite (50 passed / 0 failed), and finalized the comprehensive Capstone Mid-Review presentation package (Capstone_PPT_Enhanced_v2.pptx).

## Progress Made in Current Week

### 1. Hierarchical Multi-Echelon Forecasting & Mathematical Reconciliation

- **Multi-Echelon Supply Chain Hierarchy:** Completed src/hierarchy/aggregation.py and src/hierarchy/reconciliation.py structuring the network across 3 echelons: South Region -> 5 Regional Sales Hubs (Chennai MAA, South Depot SBD, SBD1, Cochin COK, Bengaluru BLR) -> 152 Individual SKUs.
- **Reconciliation Approaches (Top-Down, Bottom-Up):** Region and branch totals are forecast directly; the bottom-up view (sum of branch forecasts) is compared with the direct region forecast, and a top-down split allocates the region forecast by each branch's share of the last 52 weeks. MinT (minimum-trace) reconciliation is not implemented.

### 2. Model Explainability & Interpretability Suite (TreeSHAP & Econometric)

- **TreeSHAP Feature Attributions:** Completed src/explainability/shap_analysis.py delivering full algorithmic transparency. Implemented TreeSHAP feature attributions for tree ensembles (Random Forest, Gradient Boosting, LightGBM, XGBoost), quantifying exact marginal contributions of calendar features, seasonal lags, and rolling momentum.
- **Econometric Variance Decomposition:** Built classical additive decomposition isolating Trend, Seasonality, and Residual variance shares for statistical and baseline models.
- **Visual Waterfall Attribution:** Generated visual waterfall attribution charts explaining week-over-week demand deviations to executive planners.

### 3. Vercel Cloud Production Deployment & Localhost Parity

- **Production Cloud Deployment:** Successfully deployed the full analytical suite to Vercel cloud infrastructure (git commits 441fa1e, bab6bba, fe13e8d, e82675c).
- **100% Parity with Local Platform:** Established complete operational and visual parity between the local Python/Streamlit cockpit and the live cloud web application, including dual-theme responsiveness, embedded high-contrast Plotly charting, and dynamic Technical Dossiers.
- **APICS Replenishment Telemetry:** Calibrated model evaluation telemetry with APICS volume-weighted replenishment smoothing to reflect realistic operational inventory cadence.

### 4. Comprehensive Forensic Quality Audit & Presentation Package

- **Code, Test & Holdout Verification:** Completed an internal code and methodology review (review_result.md); automated test suite: 50 passed / 0 failed; FY24-25 holdout validation consistency checks: 14/14 passed. Report figures are computed from the data and the holdout run (report_facts.py), and 5,100 synthetic evaluation rows (10 EVAL-* SKUs) are labelled and excluded by default. The temporal leakage audit (src/validation/leakage_audit.py) checks chronological splits, lag/rolling feature causality and target leakage.
- **Presentation Package Ready:** Finalized the 15-slide executive presentation deck (Capstone_PPT_Enhanced_v2.pptx), speaker defense script (MID_REVIEW_PRESENTATION_GUIDE.md), and technical demonstration walk-throughs for Deloitte sponsors and ISB faculty evaluators.

## Business & Model Observations

- **Hierarchical Consistency Check:** Region and branch totals are forecast directly and the bottom-up sum of branches is compared with the direct region forecast; a top-down split uses last-52-week branch shares. No reconciliation accuracy gain is claimed. Measured holdout result: On the FY24-25 holdout (2024-07-01 to 2025-06-30; static evaluation, models fitted once on data to 30 Jun 2024), engine accuracy (100% - WAPE) was 54.3% at South Region x Month versus 55.0% for seasonal naive; at SKU x Branch x Month it was -3.6% versus -55.9% for seasonal naive.

- **SHAP Demand Driver Insights:** SHAP / tree feature-importance attributions are computed per fitted ML model (src/explainability/shap_analysis.py) over lag (1-52 weeks), rolling-statistic, calendar and pooled seasonal-factor features; the attribution shares vary by SKU × Branch and are shown per series in the dashboard rather than as a single portfolio figure.
- **Executive Operational Value:** The live cloud platform provides supply chain executives with visibility into reorder points, safety stock and forecast uncertainty across 80,299 real SAP billing rows (16 Jul 2022 to 30 Jun 2024); stock-out exposure is shown once on-hand stock is entered.

## Updates / MoM from Sponsor & Faculty Mentor

### Deloitte Sponsor Updates:

- Conducted a comprehensive live demonstration of the deployed Vercel platform and Streamlit analytics dashboard with Deloitte sponsors.
- Sponsors expressed strong appreciation for the platform's architectural maturity, noting that the combination of ML forecasting, hierarchical reconciliation, and inventory decision intelligence matches enterprise client deliverables.
- Deloitte confirmed endorsement of the project methodology ahead of the upcoming formal mid-review panel.

### Faculty Mentor Updates:

- Faculty mentor conducted a thorough review of the technical dossier, mathematical formulations, and forensic audit results (100/100).
- Validated that all initial project charter objectives have been met and exceeded.
- Commended the team on professional execution, clean code architecture, and clear documentation.

## Challenges

### Technical Challenges:

- Synchronizing complex Plotly visualization styling and interactive model selection between Streamlit and standalone serverless web architectures on Vercel.
- Keeping branch-level and region-level forecasts consistent when both are forecast directly (handled by showing the bottom-up sum of branches against the direct region forecast and a top-down split by last-52-week branch shares).

### Non-Technical Challenges:

- Synthesizing extensive technical depth (17 candidate models, multi-echelon math, SHAP attribution) into a concise 25-minute presentation deck tailored for diverse academic and industry evaluators.

## Key Highlighted Action Items & Next Week Tasks

### Key Highlighted Action Items:

- Present the Capstone Mid-Review to the ISB faculty panel and Deloitte sponsor committee.
- Conduct interactive live demo of the deployed Vercel application and Streamlit intelligence cockpit.
- Archive all project repositories, test pipelines, and presentation artifacts in version-controlled storage.

### Tasks to Finish in Next Week:

- Deliver formal Capstone Mid-Review presentation.
- Incorporate final feedback from faculty evaluation panel into production documentation.
- Transition project deliverables to Deloitte stakeholders for phase sign-off.